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Anti-oxidant effects of herbal residue from Shengxuebao mixture on heat-stressed New Zealand rabbits

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A R T I C L E I N F O

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A B S T R A C T

Heat stress can lead to hormonal imbalances, weakened immune system, increased metabolic pressure on the liver, and ultimately higher animal mortality rates. This not only seriously impairs the welfare status of animals, but also causes significant economic losses to the livestock industry. Due to its rich residual bioactive components and good safety characteristics, traditional Chinese medicine (TCM) residue is expected to become a high-quality feed additive with anti-oxidative stress alleviating function. This study focuses on the potential of Shengxuebao mixture herbal residue (SXBR) as an anti-heat stress feed additive. Through the UPLC (ultra performance liquid chromatography) technology, the average residue rate of main active ingredients from SXBR were found to be 25.39%. SXBR were then added into the basal diet of heat stressed New Zealand rabbits at the rates of 5% (SXBRl), 10% (SXBRm) and 20% (SXBRh). Heat stress significantly decreased the weight gain, as well as increased neck and ear temperature, drip loss in meat, inflammation and oxidative stress. Also, the hormone levels were disrupted, with a significant increase in serum levels of CA, COR and INS. After the consumption of SXBR in the basal diet for 3 weeks, the weight of New Zealand rabbits increased significantly, and the SXBRh group restored the redness value of the meat to a similar level as the control group. Furthermore, the serum levels T3 thyroid hormone in the SXBRh group and T4 thyroid hormone in the SXBRm group increased significantly, the SXBRh group showed a significant restoration in inflammation markers (IL-1β, IL-6, and TNF-α) and oxidative stress markers (total antioxidant capacity, HSP-70, MDA, and ROS) levels. Moreover, the real-time fluorescence quantitative PCR analysis found that, the expression levels of antioxidant genes such as Nrf2, HO-1, NQO1, and GPX1 were significantly upregulated in the SXBRh group, and the expression level of the Keap1 gene was significantly downregulated. Additionally, the SXBRm group showed significant upregulation in the expression levels of HO-1 and NQO1 genes. Western blot experiments further confirmed the up-regulation of Nrf2, Ho-1 and NQO1 proteins. This study provides a strategy for the utilization of SXBR and is of great significance for the green recycling of the TCM residues, improving the development of animal husbandry and animal welfare.

1. Introduction

Homeothermic animals have a specific temperature range known as the heat neutral zone, which typically ranges from 4 to 25 °C (Alnaimy et al., 2020). When environmental humidity and temperature exceed the thermoneutral zone, the core body temperature of animals rises,

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disrupting heat balance. This leads to various heat stress responses, such as increased respiratory rate, metabolic disorders, and elevated levels of reactive oxygen species (Aderao et al., 2023). Consequently, animals experience decreased feed intake, increased liver load, elevated inflammation levels, and higher mortality rates (Al-Sagan et al., 2020; Goma et al., 2022). It is estimated that heat stress can cost billions of dollars in economic losses worldwide every year (Wankar et al., 2021). With global climate change and the increasing occurrence of extreme weather events, the livestock industry is facing significant challenges. Many companies tend to add antibiotics to animal feed to enhance their antioxidant stress resistance. However, the misuse of antibiotics poses a hidden risk to the safety of livestock products. Prolonged consumption of various residues of antibiotics in meat contributes to the development of antibiotic-resistant bacteria, thereby endangering human health (Shao et al., 2021; Yang et al., 2021). In recent years, the development of "substitute antibiotic" feed has gained widespread interest from governments, scholars, and livestock companies (Gul, 2020; Sayyar et al., 2021; Widowati et al., 2022). For instance, Scholars have found that the addition of antioxidants to the feed of heat-stressed cattle improves their metabolism (Abeyta et al., 2023). Similarly, researchers have observed significant improvements in the antioxidant capacity and intestinal microbial structure of heat-stressed broiler chickens when adding a standardized extract of salicin from Salix alba bark as an antioxidant to their feed (Mihaela et al., 2023). Additionally, El-Gindy et al. (2023) discovered that fresh onion juice significantly enhances rabbit sperm quality and semen antioxidant status.

Shengxuebao mixture (SXB), also known as Shengxuebao Heji or Jian De, is a traditional Chinese medicine formulated from several medicinal herbs including *POLYGONI MULTIFLORI RADIX PRAEPARATA*, *LIGUSTRI LUCIDI FRUCTUS*, *MORI FRUCTUS*, *ECLIPTAE HERBA*, *PAEONIAE RADIX ALBA*, *ASTRAGALI RADIX*, and *CIBOTII RHIZOMA*. It is classified as a nourishing tonic medicine and is known for its effects in nourishing the liver and kidneys, as well as replenishing energy and blood. In clinical practice, it can be used to treat various types of anemia caused by insufficient iron intake, malnutrition, aging, and other factors (Li et al., 2022; Zhou et al., 2021). After the extraction process by pharmaceutical companies, a large amount of herbal residue (SXBR) is generated. Traditional methods of disposal such as piling up or incineration not only cause severe environmental pollution but also result in the wastage of significant amounts of active ingredients remaining in the residues (Liu et al., 2022; Lu et al., 2021). SXBR is generally recognized as safe, and its active components, such as flavonoids, terpenes, and amino acids, have the potential to enhance animals' ability to resist heat stress. There is a promising opportunity to develop SXBR as a high-quality "substitute antibiotic" feed additive (Pan et al., 2020).

In this study, the residual rates of the main components of SXB in SXBR were calculated using UPLC (ultra performance liquid chromatography) as a means to evaluate the potential of SXBR as a high-quality "substitute antibiotic" feed. Following that, 36 weaned New Zealand rabbits were allocated into a blank control group, heat stress model group, composite vitamin group, and groups with the addition of SXBR at 5%, 10%, and 20% ($n = 6$). After a 7-day adaptation period, except for the blank control group, the other groups were subjected to a 14-day heat stressing period. Performance and health indices such as average daily weight gain, feed-to-gain ratio, respiratory rate, heart rate, neck temperature, and ear temperature were evaluated. Hormone levels including CA (catecholamine), COR (cortisol), INS (insulin), T3 (triiodothyronine), and T4 (thyroxine) in the rabbits' bodies were characterized. Inflammatory levels were assessed using serum IL-1β (interleukin-1β), IL-6 (interleukin-6), and TNF-α (tumor necrosis factor-α). Furthermore, serum levels of T-AOC (total antioxidant capacity), GPX (glutathione peroxidase), SOD (superoxide dismutase), ROS (reactive oxygen species), MDA (malondialdehyde), HSP-70 (heat shock protein 70), LA (lactic acid), and PA (pyruvic acid), as well as expression levels of Nrf2 (nuclear factor erythroid2-related factor 2), HO-1 (heme Oxygenase-1), NQO1 (NADH quinone oxidoreductase 1), and Keap1 (kelch-like ECH-associated protein 1) genes and proteins in the liver, were measured using serum assay kits, real-time fluorescence quantitative PCR (polymerase chain reaction) technology, and protein immunoblotting. These measurements aimed to characterize the antioxidant stress capability of each group.

2. Materials and methods

2.1. Instruments and materials

Fluorescence quantitative PCR (CFX, Bio rad); Electrophoresis apparatus (164–5050, Bio rad); Feed granulator (YZ-160, Anhui Yangzi Industrial Technology Co., Ltd.); PH meter (S210, Mettler Toledo Instruments Co., Ltd.); Bench spectrophotometer (YS6060, Shenzhen Sanenshi Technology Co., Ltd.).

SXBR (batch 2021.09) was provided by Tsing Hua De Ren Xi'an Happiness Pharmaceutical Co., Ltd, which was prepared from 7 herbal medicines by 2 times of boil water extraction followed by drying; Dry alfalfa and concentrated feed (PK0807T3, UK Supreme Selective) were purchased from local market; Serum Reactive Oxygen Species (ROS) Test Kit (20220905, Beijing Biolab Technology Co., Ltd.); Rabbit heat shock protein 70 (HSP-70) ELISA kit (20220726, Shanghai Enzyme-linked Biotechnology Co., Ltd.); RT First Strand cDNA Synthesis Kit (G3330, Wuhan Servicebio Technology Co., Ltd.); two × SYBR Green qPCR Master Mix (None ROX) (G3320, Wuhan Servicebio Technology Co., Ltd.); NRF2, NFE2L2 Monoclonal antibody (66504-1 Ig, Wuhan Servicebio Technology Co., Ltd.); NQO1 Monoclonal antibody (67240-1-Ig, Wuhan Servicebio Technology Co., Ltd.); NQO1 Monoclonal antibody (67240-1-Ig, Wuhan Sanying Biotechnology Co., Ltd.); KEAP1 Monoclonal antibody (60027-1-Ig, Wuhan Sanying Biotechnology Co., Ltd.); HO-1/HMOX1 Monoclonal antibody (66743-1 Ig, Wuhan Sanying Biotechnology Co., Ltd.).

2.2. The determination of residual component rates

2.2.1. Preparation of test samples

Seven single herbal ingredients of SXB were powdered. According to the 2020 edition of the Chinese Pharmacopoeia, the proportions of each single herbal ingredient in the prescription of SXB were weighed accordingly. Two portions of the full formula sample of SXB were prepared in appropriate amounts. One portion was added to methanol and subjected to ultrasound extraction for 40 min, followed by filtration through a 0.22 μm microporous membrane to prepare the SXB test sample with a concentration of 50 mg/ml. The other portion was extracted with water according to the quality standards of SXB. The resulting herbal residue was dried, and an equal volume of methanol was added to prepare the SXBR test sample, following the same method as the SXB test sample. Three parallel samples were prepared for both SXB and SXBR.

2.2.2. Liquid chromatography conditions

ACQUITY UPLC HSS T3 column was used with a mobile phase consisting of 0.1% formic acid water (A) and acetonitrile (B). The elution was performed under gradient conditions as follows: 0–25 min, 2%–10% B; 25–75 min, 10%–24.5% B; 75–77 min, 24.5%–40% B; 77–80 min, 40% B; 80–90 min, 40%–2% B; 90–92 min, 2% B. The injection volume was 5 μl, the flow rate was 0.3 ml/min, and the detection wavelength was set at 254 nm.

2.2.3. Mass spectrometry conditions

AB SCIEX 5600+ QTOF high-performance liquid chromatography-triple quadrupole high-resolution mass spectrometer was used. The instrument employed an electrospray ionization (ESI) source, operated in both positive and negative ion modes. Information-dependent acquisition (IDA), dynamic background subtraction (DBS) with a 30-s interval, and high-sensitivity mode were utilized for data collection. PeakView

2.2 software was employed, with the AB SCIEX MasterView 1.1.0.0 Chinese herbal component database used as the component matching library.

2.2.4. Calculation of major component residual rates

The signal peaks with the top ten peak areas in the chromatogram of the SXB sample were collected. By comparing the retention times, the corresponding signal peaks in the SXBR sample chromatogram were identified. The residual rates of the major components of the SXB in SXBR were calculated based on the ratio of peak areas between the two samples.

2.3. Animal experiment design

Due to the fact that SXBR is a waste residue derived from herbaceous or shrubby plants, herbivorous New Zealand rabbits were selected as the experimental animals in this study. The pre-mixed concentrated feed, alfalfa grass, and SXBR were individually ground into coarse powder. They were then mixed together with required proportions and processed using a feed granulator to produce cylindrical pellets with diameter of 4 mm. After dried, the pellets were stored in a cool place until use.

Thirty-six weaned female New Zealand rabbits aged 6–7 weeks were randomly allocated into the following groups: control group (Control, 70% concentrated feed + 30% alfalfa grass), model group (Model, fed the same diet as the Control group), multivitamin group (Multivitamin, 100 mg of composite vitamins per kilogram of basic feed), low proportion SXBR group (SXBRl, 5% SXBR, 95% basic feed), medium proportion SXBR group (SXBRm, 10% SXBR, 90% basic feed), and high proportion SXBR group (SXBRh, 20% SXBR, 80% basic feed). Each rabbit was individually housed in a single cage within an SPF-level animal laboratory. They were fed with 30 g of feed per day, divided into two separate doses, and provided with free access to drinking water. The nutritional composition of the basic feed was determined according to the National Standards of the People's Republic of China, and the corresponding results are presented in Table 1.

The Control group was housed separately in a non-heat-stress feeding room, while the remaining groups were housed in a heat-stress feeding room. The air conditioning temperature in both rooms was adjusted to maintain the ambient temperature at 21–22 °C and humidity at 50%–55% during a 7-day period of adaptation without heat stress. After seven days, the air conditioning temperature in the heat-stress feeding room was adjusted and manual humidification was performed to maintain the ambient temperature at 25.5–26.5 °C and humidity at 65%–70% for a 14-day period of heat stress. At the end of the experiment, blood samples were collected to perform serum assays for antioxidant and heat stress-related markers. Liver samples were collected for real-time fluorescence quantitative PCR and protein immunoblotting analyses.

2.4. Monitoring of performance and health indices

Daily records of rabbit body weight were made to calculate the average daily weight gain and feed-to-gain ratio during the heat stress period. On the first day of heat stress, the temperature of the rabbit's ear base and neck were measured three times using an infrared digital thermometer, and the average value was calculated. The chest movements within a 10-min interval were observed and recorded, and the respiratory rate was calculated by converting the number of movements within 1 min. The color of the rabbit's ears was also observed and recorded.

Table 1
Composition and nutritional level of the basic feed.

Composition	Nutrients	Content (%)
Alfalfa grass, timothy hay, soybean husk, pea powder, douban (fragment of soybean), gleditsia powder, flaxseedsoybean oil, calcium carbonate and hydrolytic yeast	Dry matter (%)	87.72
	Organic matter (%)	81.49
	Total Carbohydrate (%)	63.43
	Crude fibre (%)	18.70
	Crude protein (%)	14.26
	Crude fat (%)	3.80
	Energy (kJ/100 g)	1461.33

2.5. Meat quality assessment

Immediately after slaughter, collected approximately 3.0 g of the longest back loin muscle from the rabbit, accurately weighed, and suspended vertically in a paper cup using string. After sealing the sample in a plastic bag, it was refrigerated at 4 °C for 24 h. The liquid that appeared on the surface of the sample was gently removed using blotting paper, and the sample was weighed again. Drip loss was calculated as the percentage of weight lost after 24 h compared to the initial weight.

At 45 min and 24 h after slaughter, approximately 1.0 g of the longest muscle from the rabbit's back loin was precisely weighed and mixed with 5.0 ml of deionized water. The pH value was measured three times using a pH meter, and the average value was recorded.

Immediately after slaughter, the longest muscle from the rabbit's back loin was used to measure the redness, yellowness, and brightness values using a meat colorimeter in a dark room. Three measurements were taken in parallel from different locations of each sample, and the average value was recorded.

2.6. Biochemical assay kit determination

Biochemical assay kits were used to measure serum levels of ROS, T-AOC, GPX, SOD, MDA, HSP-70, CA, INS, COR, T3, T4, PA, LA, TNF-α, IL-1β, and IL-6. These measurements were conducted to assess the extent of heat stress in each group and to evaluate the antioxidant function of SXBR.

2.7. Real-time fluorescence quantitative PCR was used to measure the expression of antioxidant stress-related genes in the liver

Total RNA from liver tissue was extracted using Servicebio RNA Extraction Reagent. Reverse transcription was performed using the Servicebio® RT 1st Strand cDNA Synthesis Kit to obtain cDNA templates. PCR reactions were carried out using 2 × SYBR Green qPCR Master Mix (None ROX) and the CFX fluorescence quantitative PCR instrument. The relative expression levels of the targeted gene mRNA were calculated using the $2^{-\Delta\Delta CT}$ method, with GAPDH as the reference gene. The primer sequences can be found in Table 2.

2.8. Western blot analysis was conducted to examine the expression of antioxidant stress-related proteins in the liver

The tissue lysis buffer was kept on ice, and an appropriate amount of liver tissue was weighed and finely minced on ice. The tissue-to-lysis buffer ratio was 1:10 based on the net weight of the tissue. The corresponding volume of lysis buffer was added to achieve homogenization. After centrifugation, the supernatant was collected and divided into two tubes for protein quantification and gel electrophoresis. The supernatant was mixed with 4x Laemmli sample buffer at a 1:3 ratio, depending on the protein concentration, and thoroughly combined. The mixture was then heated in boiling water for 8 min to denature the proteins. A 10% separating gel and a 5% stacking gel were prepared. Gel electrophoresis, membrane transfer, blocking, and overnight incubation with the primary antibody were performed at 4 °C. The membrane was washed three times with TBST for 5 min each time. The appropriate secondary antibody was added and incubated at room temperature for 1 h. After washing the membrane three times with TBST, it was exposed and developed for visualization.

2.9. Data statistical analysis and plotting

The data were analyzed using IBM SPSS Statistics 26, utilizing t-tests with a significance threshold set at $P < 0.05$. Bar graphs were created using GraphPad Prism 9.0.

3. Results

3.1. Residual levels of major components in SXBR

To determine the potential of SXBR as a functional feed with antioxidant stress properties, this study utilized UPLC technology to detect the component signals of SXB and SXBR. The results were matched with the mass spectrometry identification results, and the component mass spectra are shown in Supplementary Fig. 1. Finally, the residual rates of major components were calculated based on peak area ratios. The results indicated that the residual rates of major components in SXBR ranged from 0.00% to 52.96%, with an average residual rate of 25.39% (Table 3, Fig. 1). This suggests that SXBR contains an abundance of active components such as flavonoids, terpenes, and saponins. Developing it into feed not only reduces waste of traditional Chinese medicine resources but also has the potential for high-quality functional "substitute antibiotic" feed.

3.2. Results of performance and health indices characterization

From the results of performance and health indices characterization, it can be observed that the average daily weight gain of the rabbits in the model group was significantly lower than that of the blank control group ($P < 0.05$) (Fig. 2A). The feed-to-gain ratio, neck temperature, and ear temperature were significantly higher in the model group compared to the blank control group ($P < 0.05$) (Fig. 2BEF). The rabbits in the model group exhibited a noticeable reddening of the ears and roughening of the fur (Supplementary Fig. 2). With an increase in the proportion of SXBR addition, the average daily weight gain of the rabbits also increased, and the average daily weight gain of all SXBR groups was significantly higher than that of the model group ($P < 0.05$) (Fig. 2A). Similarly, with an increase in the proportion of SXBR addition, the feed-to-gain ratio of all SXBR groups was significantly lower than that of the model group ($P < 0.05$) (Fig. 2B). There were no significant differences in respiratory rate, heart rate, neck temperature, and ear temperature between the SXBR groups and the model group (Fig. 2CDEF). These results indicate that SXBR can regulate weight gain in animals under heat stress and alleviate the negative impact of heat stress on livestock productivity.

3.3. Effects of SXBR on the meat quality of New Zealand rabbits under heat stress conditions

Heat stress can severely damage the meat quality of livestock by disrupting fat, carbohydrate, and lactate metabolism, resulting in economic losses. The results of meat quality assessment showed that the drip loss rate in the model group was significantly higher than that in the blank control group ($P < 0.05$) (Fig. 3F). Although there was no statistically significant difference in the meat redness values between the model group and the control group, the meat redness values of the SXBRh group and the multivitamin group were significantly higher than those of the model group ($P < 0.05$) and similar to the control group (Fig. 3A). Although there were no significant differences in yellowness value, brightness value, and drip loss rate among the groups, these values decreased with an increase in the proportion of SXBR addition (Fig. 3BCF). There were no significant differences in meat pH among the groups (Fig. 3DE). These findings suggest that SXBR can improve meat quality to some extent and reduce production losses.

Table 2
Primer sequence of targeted gene.

Name	Accession number	Primer sequence (5′-3′)	length (bp)
Nrf2	XM_002712305.3	GATGGACTTGGAGCTACCGC ACGTCTCGACTTACCCCAAG	105
HO-1	XM_002711415.3	GGTGACTGCCGAGGGTTTTA GTTGTGCTCAATCTCCTCCTCC	82
NQO1	XM_017342326.1	CAGGAAGGACATCACAGGCAA CTTCTGAAATATCACCAGGTCTGC	151
Keap1	XM_008251549.2	GATGCTGACCCGAAGGATCG TGATCATCCGCCACTCGTTC	137
SOD1	NM_001082627.2	GGTGGTCAAGGGACGCATAA AGTCACATTACCCAGGTCGC	178
GPX1	NM_001085444.1	ACTACGGTCCGGGACTACAC TCTTGGCGTTCTCCTGATGC	115
GAPDH	NM_001082253.1	TTCCAGTATGATTCCACCCACG GGGCTGAGATGATGACCCTTTT	232

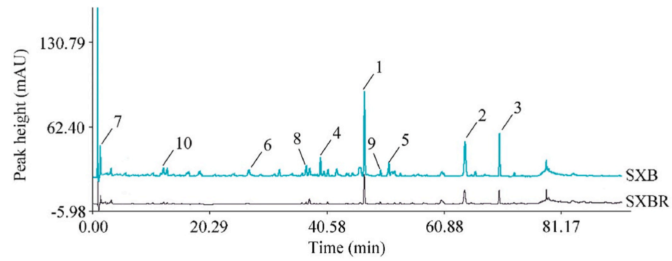

Fig. 1. UPLC chromatograms of SXB and SXBR.
Note: In the figure, the cyan color represents the chromatogram of SXB, the black color represents the chromatogram of SXBR, and numbers 1–10 represent the top ten signal peaks in SXB in terms of intensity.

Table 3
Residual levels of major components of SXB in SXBR.

Number	Name	Mass	Peak area		Residual rate %
			SXB (mAU*s)	SXBR (mAU*s)	
1	Calycosin	284.07	758.3751	278.0964	36.67
2	Ecliptasaponin A	634.41	478.1495	137.4334	28.74
3	Linoleic acid	280.24	379.5115	126.6481	33.37
4	Specnuezhenide	686.24	176.2708	63.6355	36.10
5	Astragaloside A	830.47	157.3779	21.1830	13.46
6	Calycosin-7-glucoside	446.12	137.9361	0.0000	0.00
7	Protocatechuic aldehyde	138.03	124.8787	66.1347	52.96
8	Luteoloside	448.10	112.6459	17.9284	15.92
9	Formononetin	268.07	110.5724	21.3218	19.28
10	Paeoniflorin	480.16	91.6117	15.9017	17.36

3.4. Effects of SXBR on hormone levels in heat-stressed New Zealand rabbits

Heat stress can lead to increased levels of catecholamines and glucocorticoids, as well as decreased levels of insulin and thyroid hormones. In this study, the hormone levels were measured in different groups of heat-stressed New Zealand rabbits. The results showed that the levels of CA, COR and INS were significantly elevated in the model group compared to the blank control group ($P < 0.05$) (Fig. 4ABC). The levels of T3 and T4 were slightly decreased in the model group but did not show significant differences compared to the blank control group. The SXBRh group exhibited a significant increase in T3 level compared to the model group, while the SXBRm group showed significantly higher T4 level than the model group ($P < 0.05$) (Fig. 4DE). Furthermore, with an increasing proportion of SXBR, there is a gradual restoration in the levels of COR, INS, and T3. These results indicate that SXBR can regulate endocrine disruption caused by heat stress in New Zealand rabbits, maintaining their overall health.

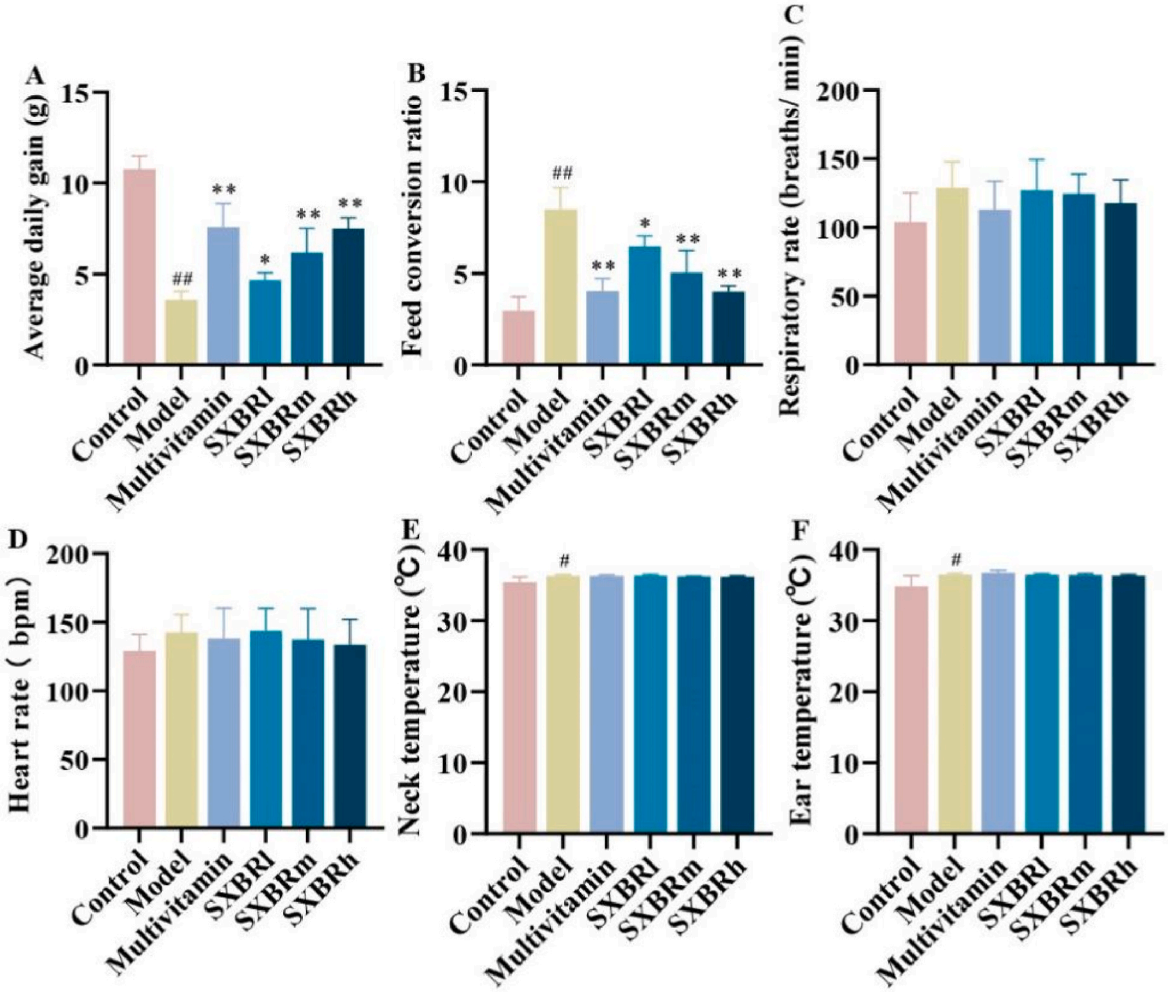

Fig. 2. Monitoring results of growth and physiological status in different groups of New Zealand rabbits.
Note: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

3.5. Effects of SXBR on serum inflammatory cytokine levels in heat-stressed New Zealand rabbits

Comparing the serum inflammatory cytokine levels between the model group and the blank control group, it can be observed that heat stress significantly increased the levels of TNF-α, IL-1β, and IL-6 in New Zealand rabbits ($P < 0.05$). The SXBRl and SXBRm groups showed no significant differences in inflammatory levels compared to the model group. However, the SXBRh group exhibited significantly lower levels of TNF-α, IL-1β, and IL-6 in the serum compared to the model group ($P < 0.05$) (Fig. 5). These results indicate that SXBR at the rate of 20% has a notable anti-inflammatory effect and can effectively reduce the inflammatory levels in animals under heat stress conditions.

3.6. Characterization of the antioxidant performance of SXBR through serum assay kit detection

The results of serum antioxidant index detection showed that the GPX activity in the model group was significantly lower than that in the blank control group ($P < 0.05$). Additionally, the levels of MDA, ROS, HSP-70, and lactate (LA) in the serum were significantly higher in the model group compared to the blank control group ($P < 0.05$) (Fig. 6BDEFG). The SXBRh group exhibited a significant increase in serum T-AOC and HSP-70 levels compared to the model group, while the SXBRm group showed significantly higher T-AOC level than the model group ($P < 0.05$) (Fig. 6AF). Moreover, the SXBRh group also had significantly lower levels of MDA and ROS in the serum compared to the model group ($P < 0.05$) (Fig. 6DE). There were no significant differences in GPX activity, SOD activity, LA content, and pyruvic acid (PA) content between the SXBR groups and the model group (Fig. 6BCGH). These findings indicate that SXBR exhibits good antioxidant performance based on serum antioxidant levels.

3.7. PCR detection of antioxidant gene expression in the liver of different groups of New Zealand rabbits

The liver is the main site of metabolic activity in the body. Under heat stress, there is metabolic disruption, increased production of reactive oxygen species, and increased metabolic pressure on the liver, leading to irreversible liver damage. In this study, the expression levels of liver antioxidant genes were determined using PCR. The results showed that the expression levels of HO-1, NQO1, GPX1, and SOD1 genes in the model group were significantly lower compared to the blank control group (Fig. 7BCDE). Additionally, the expression level of the Keap1 gene was significantly increased in the model group compared to the blank control group (Fig. 7D), indicating severe liver damage and decreased antioxidant stress function in the heat stress model. After adding SXBR to the feed, the SXBRh group showed significant upregulation of Nrf2, HO-1, NQO1, and GPX1 gene expression in the liver compared to the model group ($P < 0.05$) (Fig. 8ABCE). The expression level of the Keap1 gene was significantly decreased in the SXBRh group compared to the model group ($P < 0.05$) (Fig. 7D). The SXBRm group

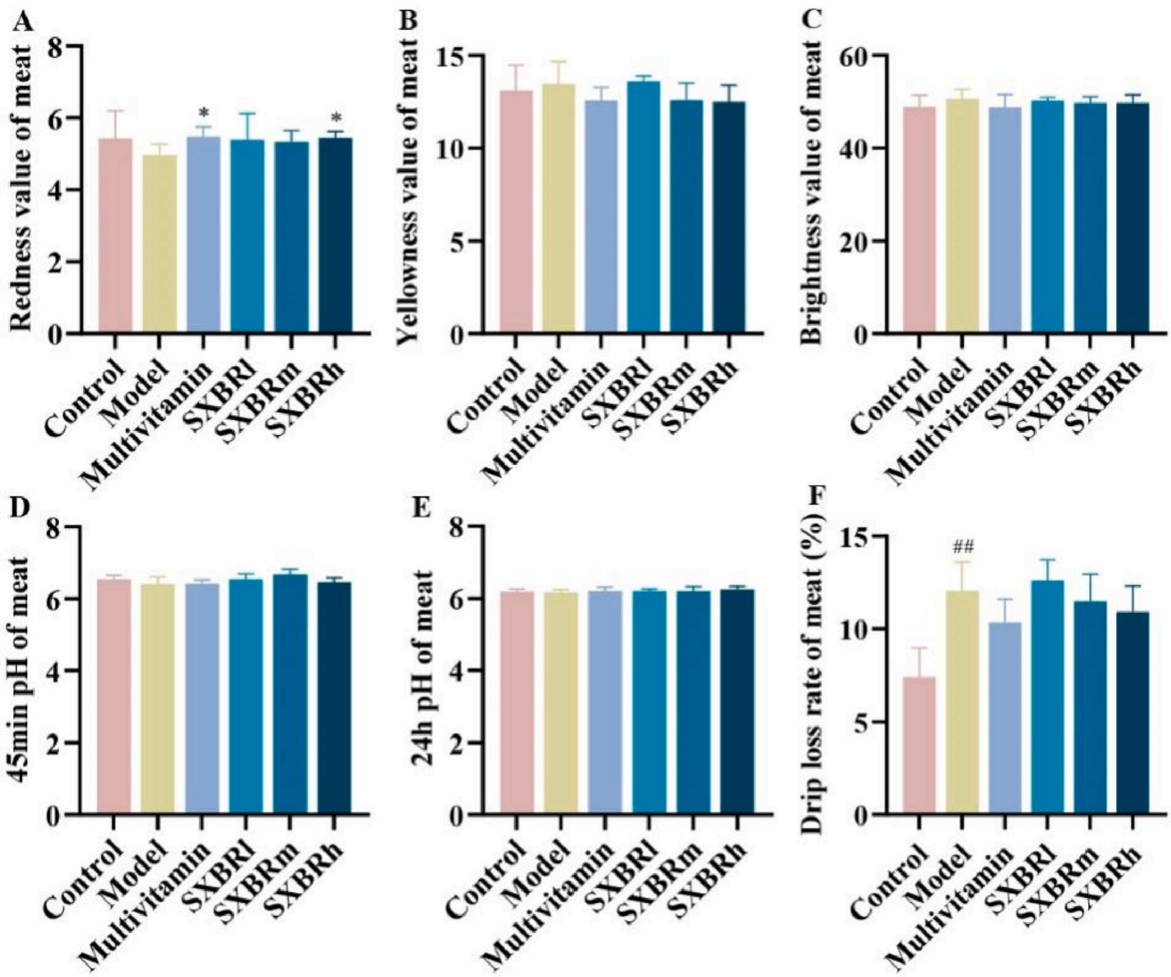

Fig. 3. Results of meat quality assessment in different groups of New Zealand rabbits.
Note: $^{\#\#}P < 0.01$, compared with Control; $^{*}P < 0.05$, compared with Model.

also exhibited significantly higher expression levels of HO-1 and NQO1 genes in the liver compared to the model group ($P < 0.05$) (Fig. 7BC). There were no significant differences in SOD1 gene expression in the liver between the SXBR treatment groups and the model group (Fig. 7F).

3.8. Western blot analysis detected liver antioxidant protein expression and predicted related pathways in New Zealand rabbits from different groups

To further validate the enhanced antioxidant stress response of the liver by SXBR, protein immunoblotting was performed to detect the expression levels of antioxidant-related proteins in the liver of New Zealand rabbits from different groups. The results were consistent with the PCR-based assessment of antioxidant gene expression. The protein expression levels of Nrf2, HO-1, and NQO1 were significantly lower in the liver of the model group compared to the blank control group. However, the addition of SXBR significantly increased the protein expression levels of Nrf2, HO-1, and NQO1 in the liver of the SXBR groups compared to the model group ($P < 0.05$) (Fig. 8ABCD). There were no significant differences in Keap1 protein expression levels among the groups (Fig. 8AE).

Based on the results obtained from serum assay kit measurements, real-time fluorescence quantitative PCR, and Western blot analysis, it can be inferred that SXBR may exert its antioxidant effects through direct or indirect clearance of ROS (Fig. 9). On one hand, SXBR may exhibit direct antioxidant activity by blocking ROS production in the liver and directly scavenging ROS. On the other hand, SXBR may activate the Nrf2/Keap1/HO-1/NQO1 antioxidant signaling pathway in the liver, promoting the expression of antioxidant-related genes and proteins, thereby clearing ROS, reducing malondialdehyde (MDA), and exerting indirect antioxidant effects.

4. Discussion

Chronic heat stress leads to increased levels of catecholamines and glucocorticoids, decreased levels of thyroid hormones, reduced weight gain, and deteriorated meat quality in animals (Habeeb, 2020; He et al., 2018). Catecholamines mediate the activation of the autonomic nervous system (ANS), leading to increased respiratory rate and heart rate, elevated body temperature, redistribution of blood from internal organs to the skin for thermoregulation, promotion of energy utilization from body reserves, acceleration of muscle glycogen breakdown, and inhibition of energy storage (Bisogni et al., 2016); The action of catecholamines on β2 receptors in muscles initiates a cascade reaction mediated by cyclic adenosine monophosphate (cAMP). This reaction activates glycogen phosphorylase and inhibits glycogen synthase, leading to the activation of muscle glycogen breakdown and inhibition of glycogen synthesis (Guilherme et al., 2023). The body increases plasma concentrations of glucocorticoids by activating the hypothalamic-pituitary-adrenal (HPA) axis. Glucocorticoids enhance heat loss through vasodilation and induce changes in protein hydrolysis and lipid metabolism. They promote the hydrolysis of circulating

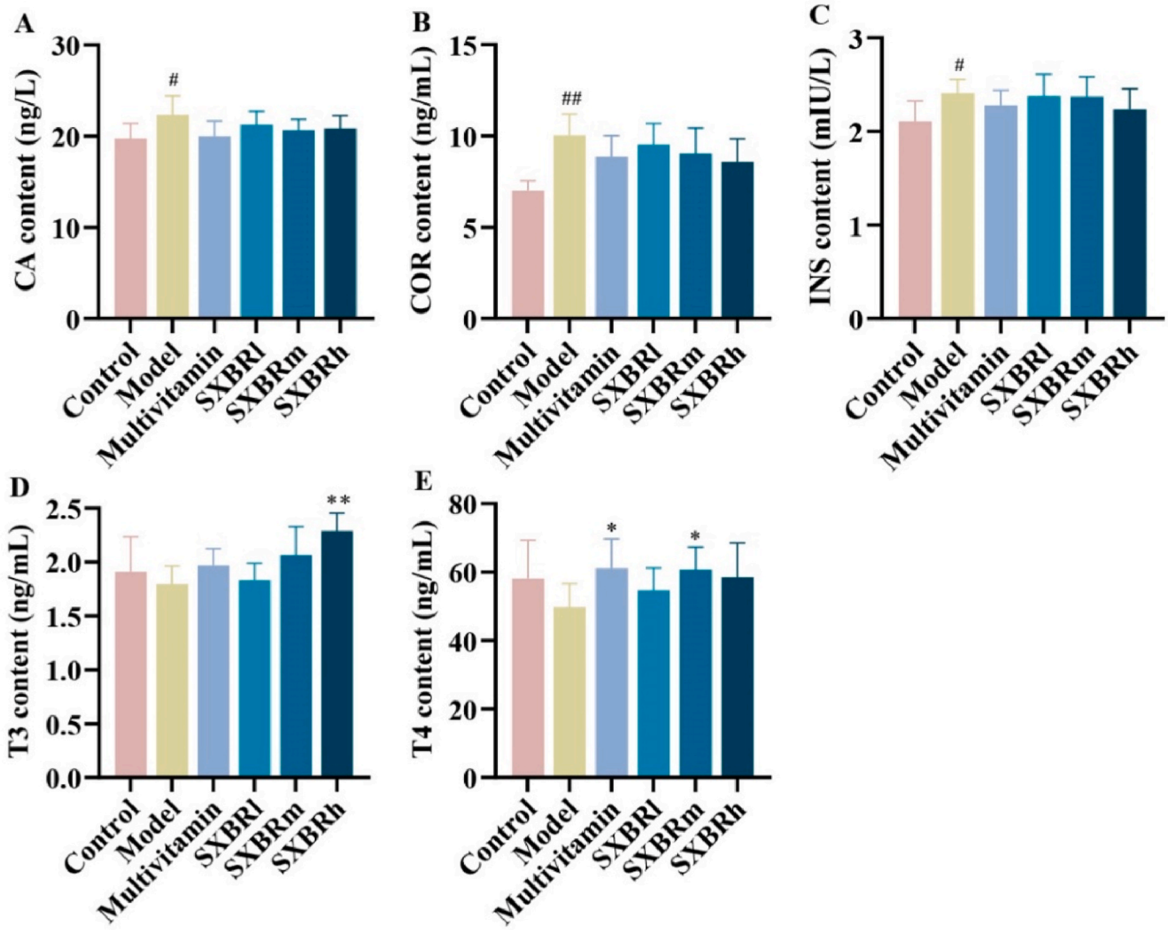

Fig. 4. Hormone levels in New Zealand rabbits in each group.
Note: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

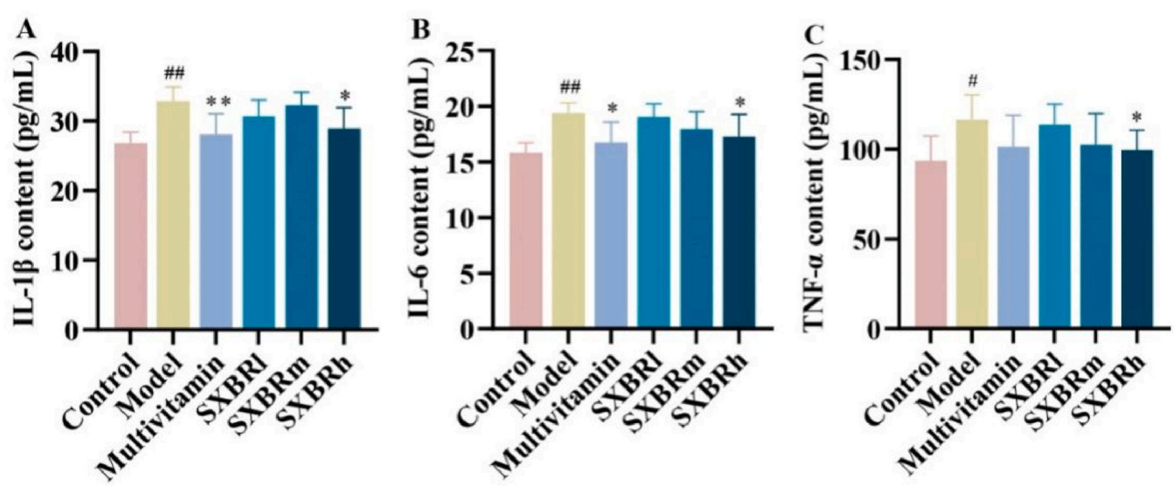

Fig. 5. Inflammatory cytokines levels in the serum of different groups of New Zealand rabbits.
Note: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

triglycerides by increasing lipoprotein lipase activity, while also inducing fat synthesis in liver cells and adipocytes by upregulating fatty acid synthase expression (Gonzalez-Rivas et al., 2020). Panting and sweating are the only ways for heat-stressed animals to dissipate heat, which leads to significant water loss. As a result, the animals' water requirements increase. Inadequate water supply to heat-stressed animals can lead to dehydration and enhanced adrenergic stress response, increased muscle glycogen utilization, and ultimately, compromised meat quality (Collier et al., 2017). Thyroid hormones are closely associated with animal thermoregulation. Heat stress typically leads to a decrease in plasma concentrations of triiodothyronine (T3) and thyroxine (T4) (Kahl et al., 2015). Furthermore, heat stress can stimulate

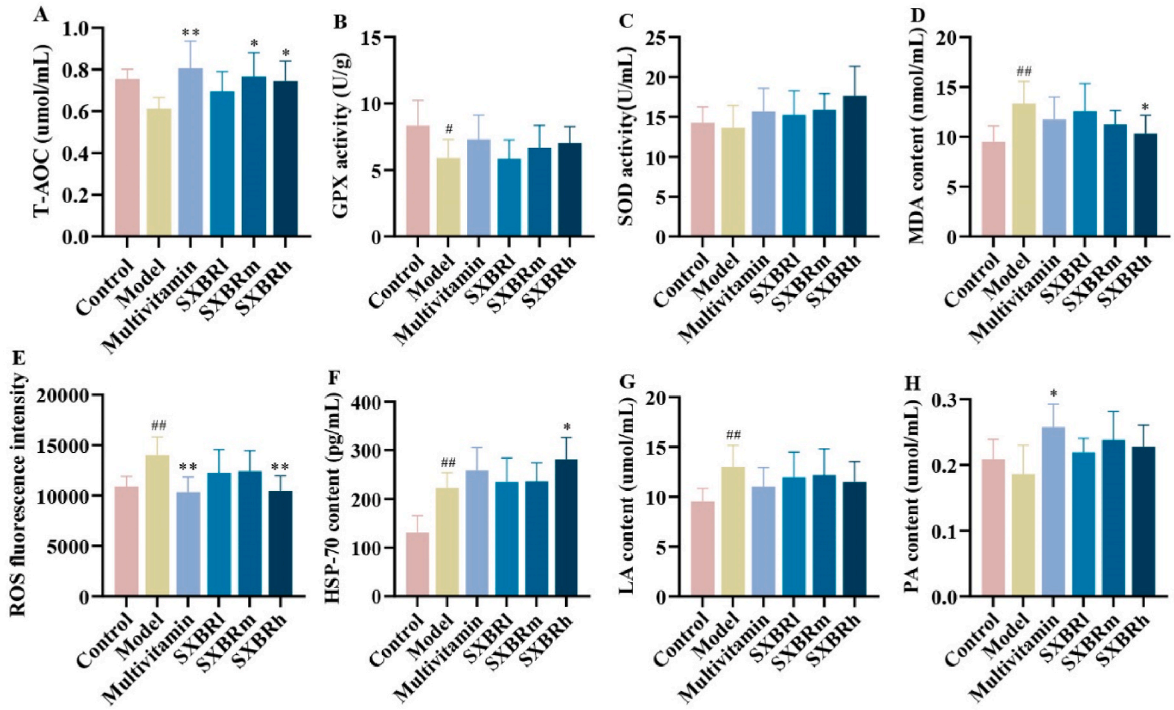

Fig. 6. Serum antioxidant levels in different groups of New Zealand rabbits.
Note: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

insulin secretion and increase systemic insulin sensitivity in animals (Sammad et al., 2020). These two responses are considered as adaptive mechanisms to reduce metabolic heat production, maintain energy requirements, and avoid additional heat load by reducing fat breakdown and increasing fat deposition. The optimal temperature for rabbit rearing is between 15 and 25 °C (Dusboevich, 2022). In this study, the rearing room temperature was set at 25.5–26.5 °C. It was observed that the weight gain of New Zealand rabbits significantly decreased, body temperature increased significantly, the drip loss in meat increased, and serum levels of catecholamines (CA) and cortisol (COR) significantly increased, while triiodothyronine (T3) and thyroxine (T4) levels slightly decreased. The hormone levels were disrupted, and inflammatory markers significantly increased, indicating the successful establishment of the heat stress model in this study. After adding SXBR, the weight gain of New Zealand rabbits improved significantly, the meat redness values recovered to a level similar to that of the control group, and INS, T3 and T4 levels showed significant recovery compared to the model group. Inflammation levels decreased significantly. However, the effect of SXBR on temperature regulation under heat stress was relatively weak.

Excessive ventilation caused by panting can lead to respiratory alkalosis, and the body compensates rapidly by excreting HCO3- in urine, resulting in a decrease in blood pH and the development of metabolic acidosis (Habibu et al., 2019). During metabolic acidosis, muscles rely more on anaerobic metabolism to produce energy, leading to the conversion of more pyruvate into lactate. This results in the occurrence of PSE-like characteristics in meat, with pale color and drip loss (Zaboli et al., 2019). Heat stress can also increase the levels of ROS and potentially reduce the natural antioxidant capacity (Klumpen et al., 2017). Metabolic acidosis and increased levels of ROS both trigger oxidative stress, resulting in liver damage (Liu et al., 2021). This is manifested by abnormal expression of antioxidant genes such as Nrf2, HO-1, NQO1, GPX1, SOD1, CAT, MT3, OVGP1, UCP1, and OXSR1 (Serra et al., 2023; Sikiru et al., 2019). Oxidative stress and tissue acidosis lead to cellular toxicity, free radical-mediated chain reactions, protein and lipid oxidation, impaired animal health and productivity, and ultimately, poor product quality with a shortened shelf life (Mavrommatis et al., 2021; Valko et al., 2016; Sindi et al., 2023). Heat stress can also lead to an adaptive increase in the expression of HSP-70 in animals (Belhadj Slimen et al., 2014). The results of this study indicate that heat stress indeed leads to increased levels of LA, ROS, MDA and HSP-70 in the serum of New Zealand rabbits. It also significantly reduces GPX activity and downregulates the gene and protein expression of Nrf2, HO-1, and NQO1 in the liver, while upregulating the gene expression of Keap1 compared to the blank control group. These findings suggest that heat stress imposes a heavy burden on the liver of New Zealand rabbits, leading to a decrease in antioxidant stress response. The regulatory effects of SXBR on respiration and heart rate were not significant, which is consistent with the measurement of LA and PA levels in the serum, as there were no significant differences between the SXBR groups and the model group. However, the addition of a high proportion of SXBR in the feed effectively reduced the serum levels of ROS and MDA in heat-stressed New Zealand rabbits, while enhancing the serum T-AOC. The gene and protein expression levels of Nrf2, HO-1, and NQO1 in the liver were significantly increased, along with a notable increase in GPX gene expression and a significant decrease in Keap1 gene expression compared to the model group. Keap1, as part of the E3 ubiquitin ligase complex, is involved in the constitutive ubiquitination and proteasome-dependent degradation of the target gene Nrf2. However, when the organism experiences an imbalance between oxidation and antioxidant defense, Keap1 gene exhibits abnormally high expression. SXBR may exert its antioxidant effects by activating the Nrf2/Keap1/HO-1/NQO1 signaling pathway. On one hand, it reduces the expression of the Keap1 gene. On the other hand, it promotes modification of Keap1's cysteine residues, altering its conformation. As a result, Nrf2 is released and translocates into the nucleus, where it forms a heterodimer with small Maf proteins and binds to the antioxidant response element (ARE), inducing the expression of antioxidant genes and proteins such as HO-1 and NQO1 (Baird et al., 2020). Additionally, with the addition of SXBR, the serum HSP-70 levels significantly increased compared to the model group. Previous research has shown that HSP-70 is significantly upregulated under heat stress to prevent cellular damage (Eroglu et al., 2014; Li et al., 2014). SXBR may

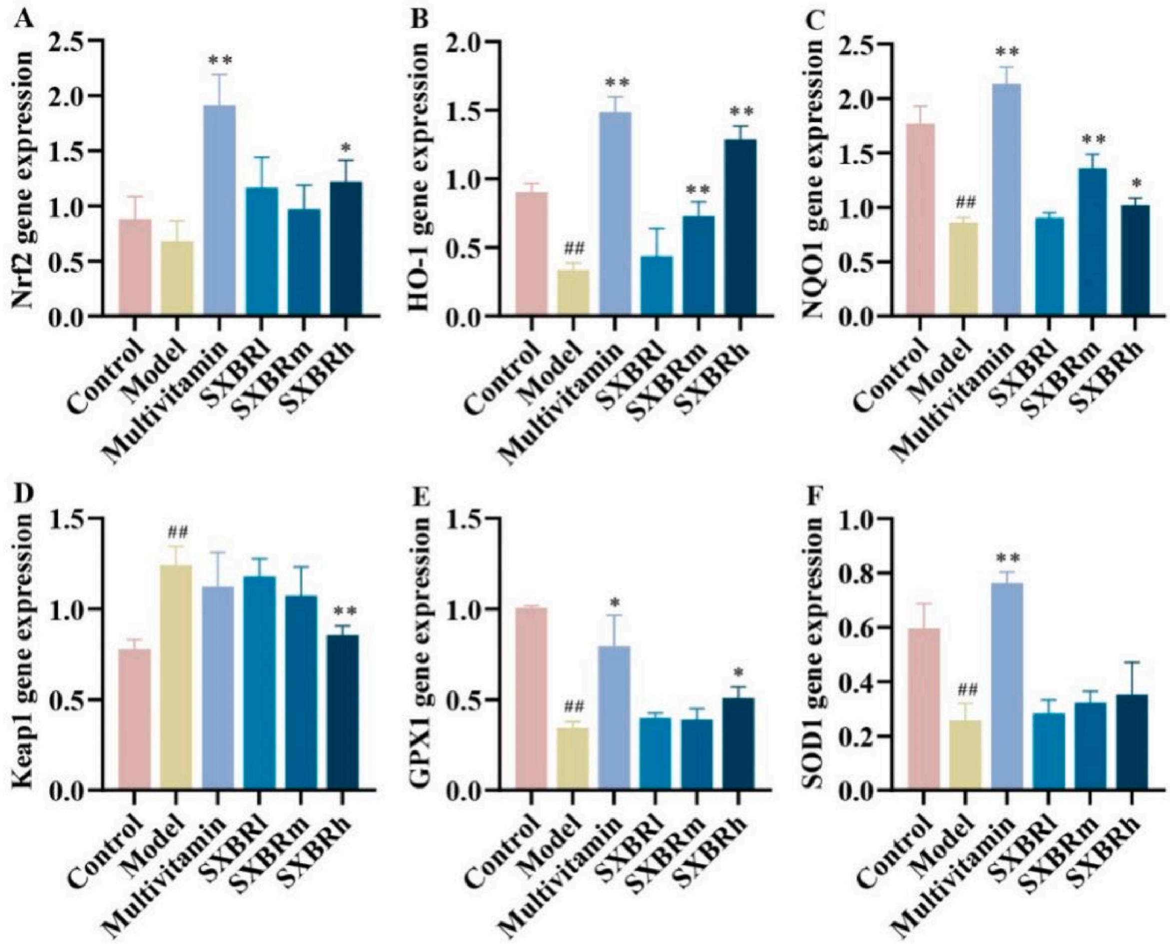

Fig. 7. Expression levels of antioxidant genes in the liver of different groups of New Zealand rabbits.
Note: $^{##}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

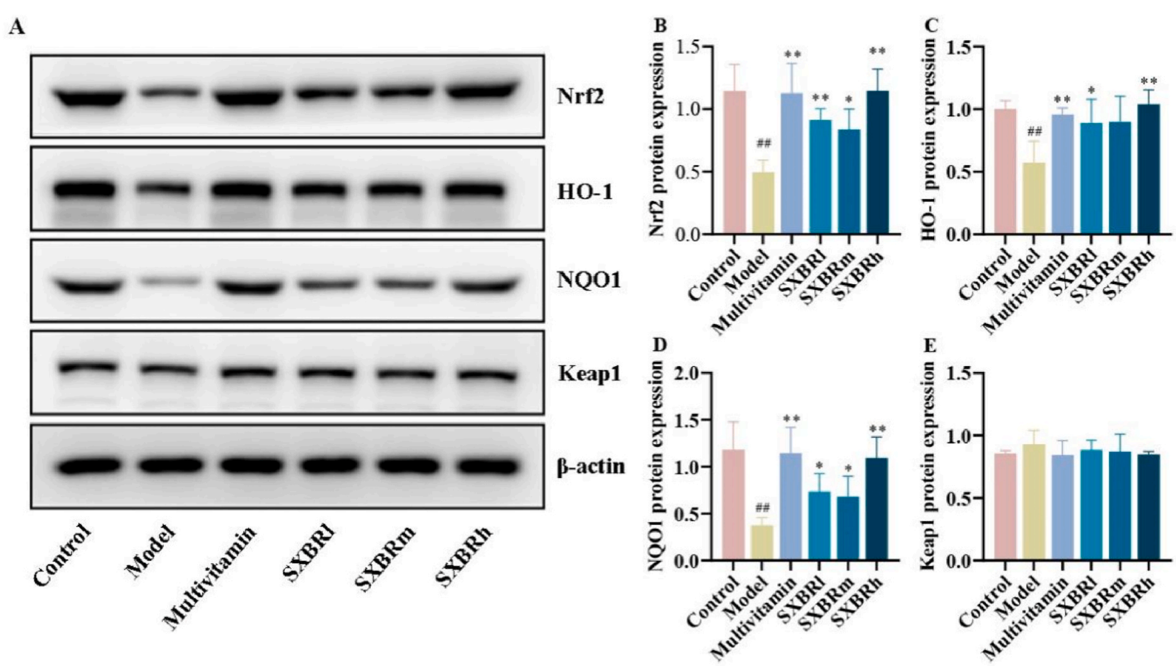

Fig. 8. Expression levels of antioxidant proteins in the liver of different groups of New Zealand rabbits.
Note: $^{##}P < 0.01$, compared with Control; $^{*}P < 0.05$, $^{**}P < 0.01$, compared with Model.

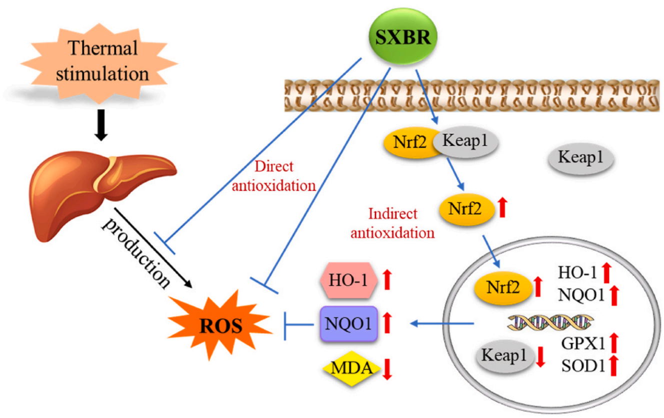

Fig. 9. Prediction of the pathways involved in SXBR's antioxidant actions.

further promote its release and enhance the protective effects of HSP-70 on cells.

Currently, feed additives used for animal heat stress mainly include electrolytes, vitamins, antibiotics, and herbal medicines (Ali et al., 2021; Liang et al., 2022). The SXBR used in this study is the residue after water extraction of seven medicinal herbs from the same source as food, which has good safety profile. It contains a variety of antioxidant active substances. Compared to excessive use of electrolyte and vitamin feed additives that can lead to animal toxicity, and antibiotic feed additives that can cause bacterial resistance and harm human health, SXBR provides a safer alternative. Moreover, compared to directly using herbal medicines in feed, SXBR helps conserve herbal resources and fully utilize their benefits.

5. Conclusion

SXBR contains approximately 25.39% residual active ingredients from a comprehensive formula, showing potential for development as a functional "anti-stress" feed with antioxidant stress capabilities. Results from animal experiments using a heat stress model demonstrated that SXBR promoted weight gain, restores the meat redness value to a higher level, regulates hormone levels, reduces inflammation, effectively enhances the antioxidant stress capacity of heat-stressed New Zealand rabbits, and prevents irreversible liver damage. This study provides reference for the development of animal feed resources and holds significant importance in promoting animal welfare and achieving the green circular utilization of SXBR.

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CRediT authorship contribution statement

Yu He: Investigation, Data Curing, Writing Original Draft Preparation, Methodology, and Visualization. **Jingao Yu:** Investigation, Data curation, Writing – original draft, Methodology, Visualization. **Zhongxing Song:** Formal analysis, Writing – review & editing. **Zhishu Tang:** Project administration. **Jin-Ao Duan:** Conceptualization, Supervision. **Huaxu Zhu:** Conceptualization, Supervision. **Hongna Liu:** Conceptualization, Supervision. **Jianping Zhou:** Resources. **Zhaojun Cao:** Resources.

Declaration of competing interest

The authors report there are no competing interests to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jtherbio.2023.103752.

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